Vietnam National University of HCMC

International University

School of Computer Science and Engineering

Program of MIT

Course: Advanced Computer Graphics (IT516)

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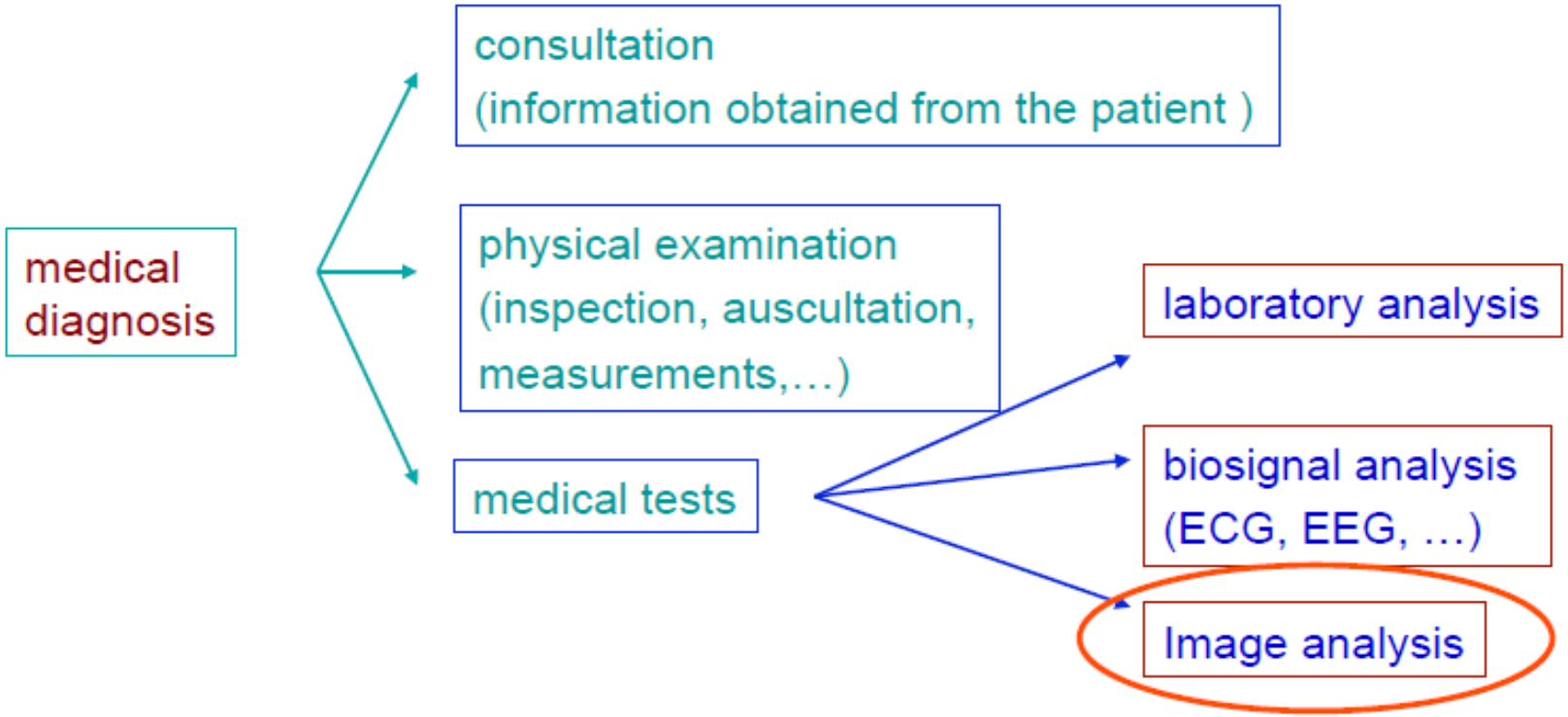
Medical image processing (MIP)

- This topic covers researches in MIP and their application in practice
- **Medical imaging** is the technique and process of creating visual representations of the interior of a body for clinical analysis and medical intervention, as well as visual representation of the function of some organs or tissues
- Libraries for MIP:
  - VTK (Visualization Toolkit): http://www.vtk.org
  - Open sources: framework &tools https://medevel.com/build-medical-imaging-apps-with-these-10-open-source-frameworks

Medical Imaging



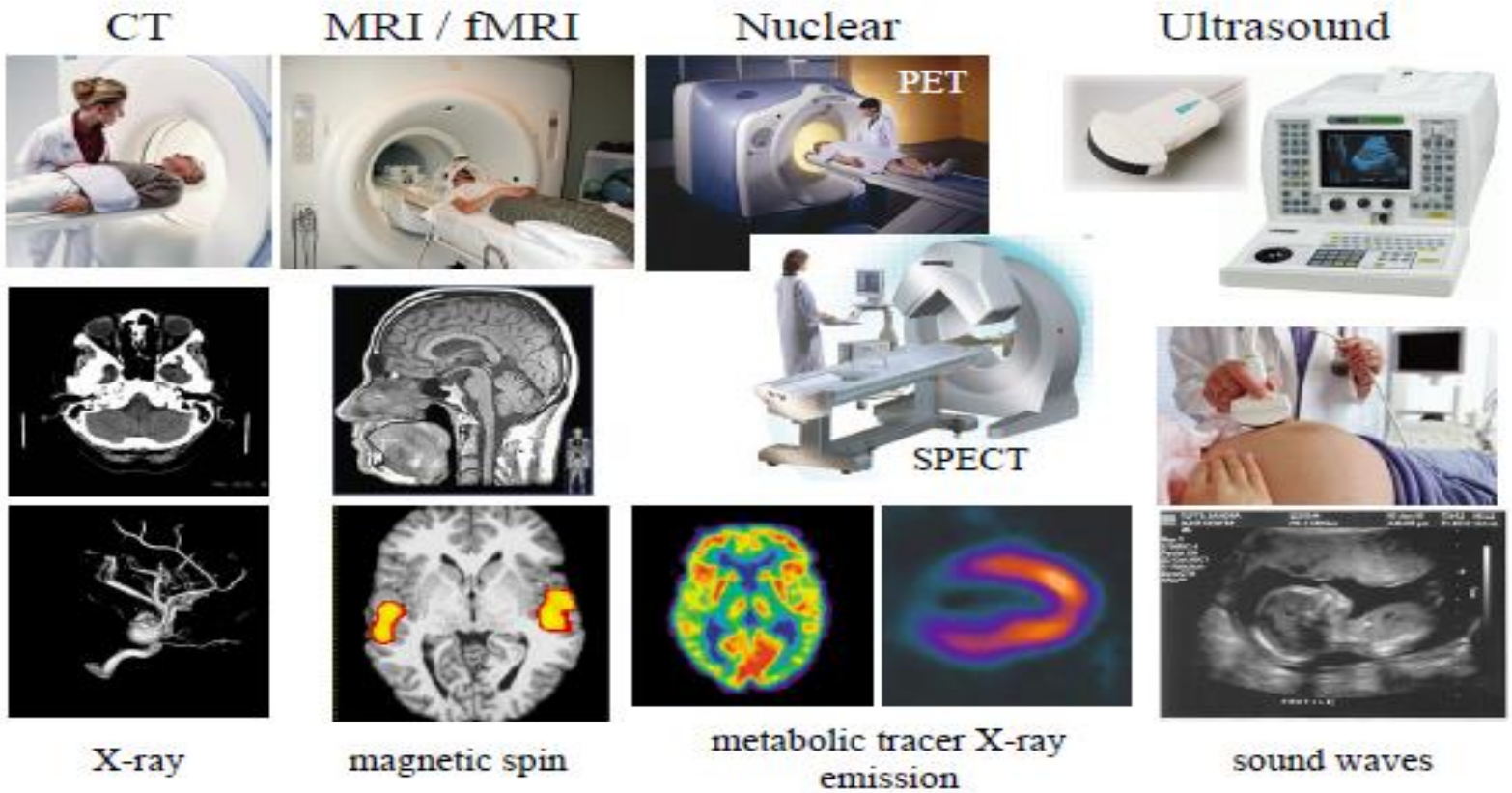
Why is Medical Imaging Important?

- To diagnose, treat and cure patients without causing any harmful side effects.
- Computer vision (combine with CG and/or GM) can exploit texture, shape, contour and prior knowledge along with contextual information from image sequence and provide 3D and 4D information that helps with better human understanding.
- To see inside a patient without having to cut them open
- The final objective is to benefit the patients without adding to the already high medical costs.

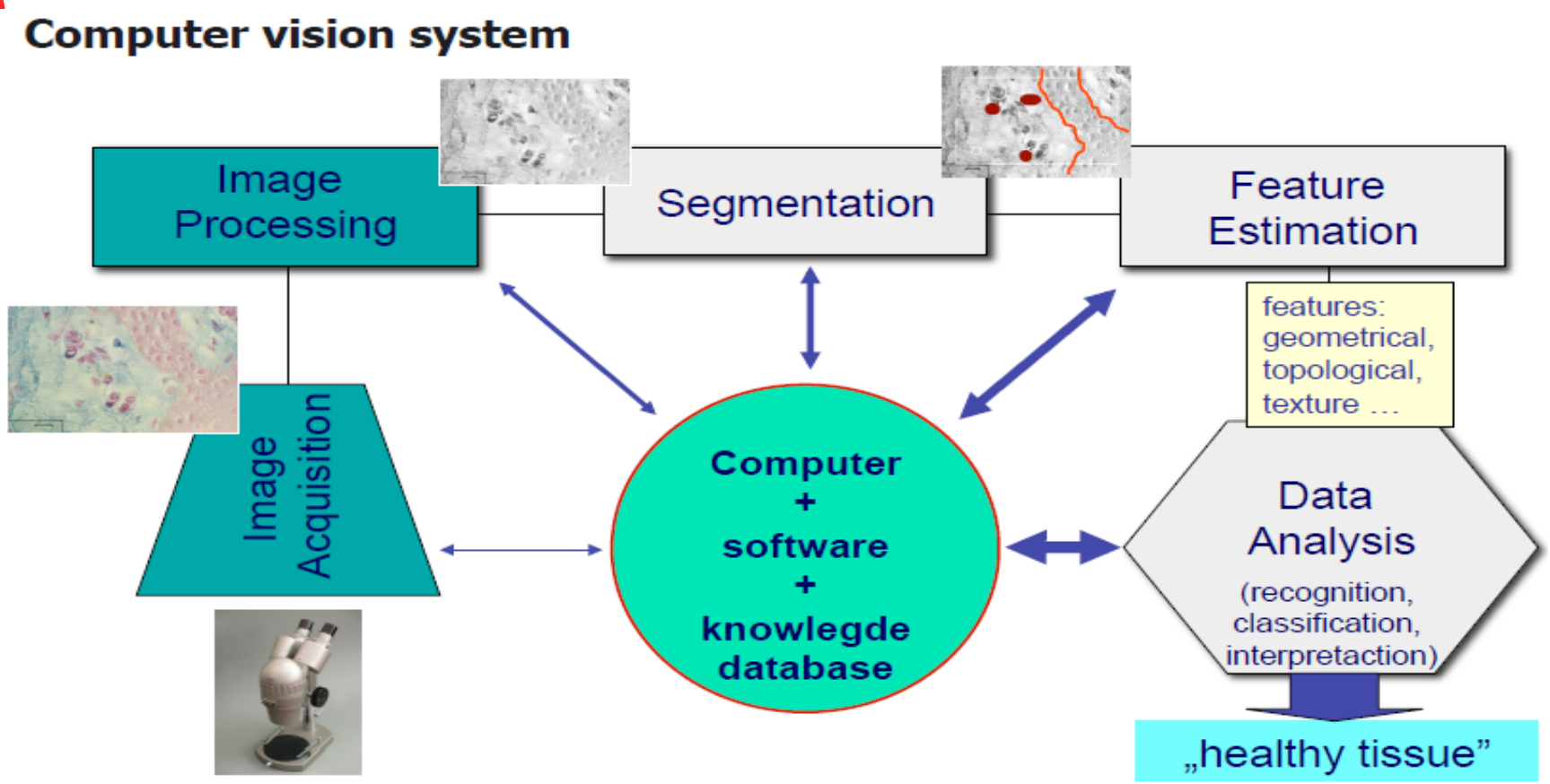
Image Modalities

- X-ray
- Computed Tomography (CT)
- Magnetic Resonance Imaging (MRI)
- Single Photon Emission Computed Tomography (SPECT)
- Positron Emission Tomography (PET)
- Ultrasound
- Nuclear Magnetic Resonance (NMR)

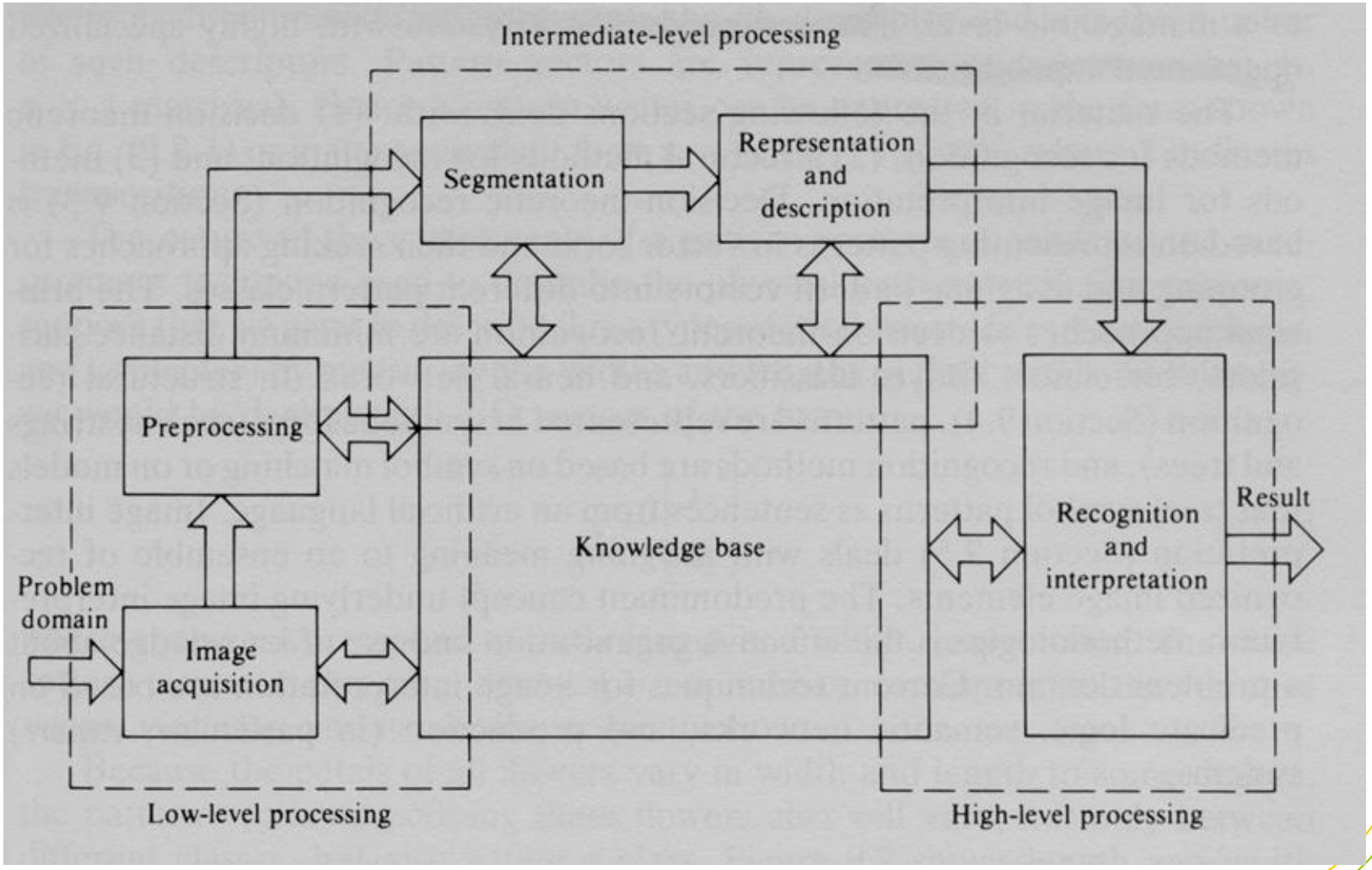
Different Image Modalities



Computer Vision System



Levels of Processing



Levels of Processing

- Low-level processing
  - Standard procedures are applied to improve image quality
  - Procedures are required to have no intelligent capabilities.
- Intermediate-level processing
  - Extract and characterize components in the image
  - Some intelligent capabilities are required.
- High-level processing
  - Recognition and interpretation.
  - Procedures require high intelligent capabilities.

Architecture Modules

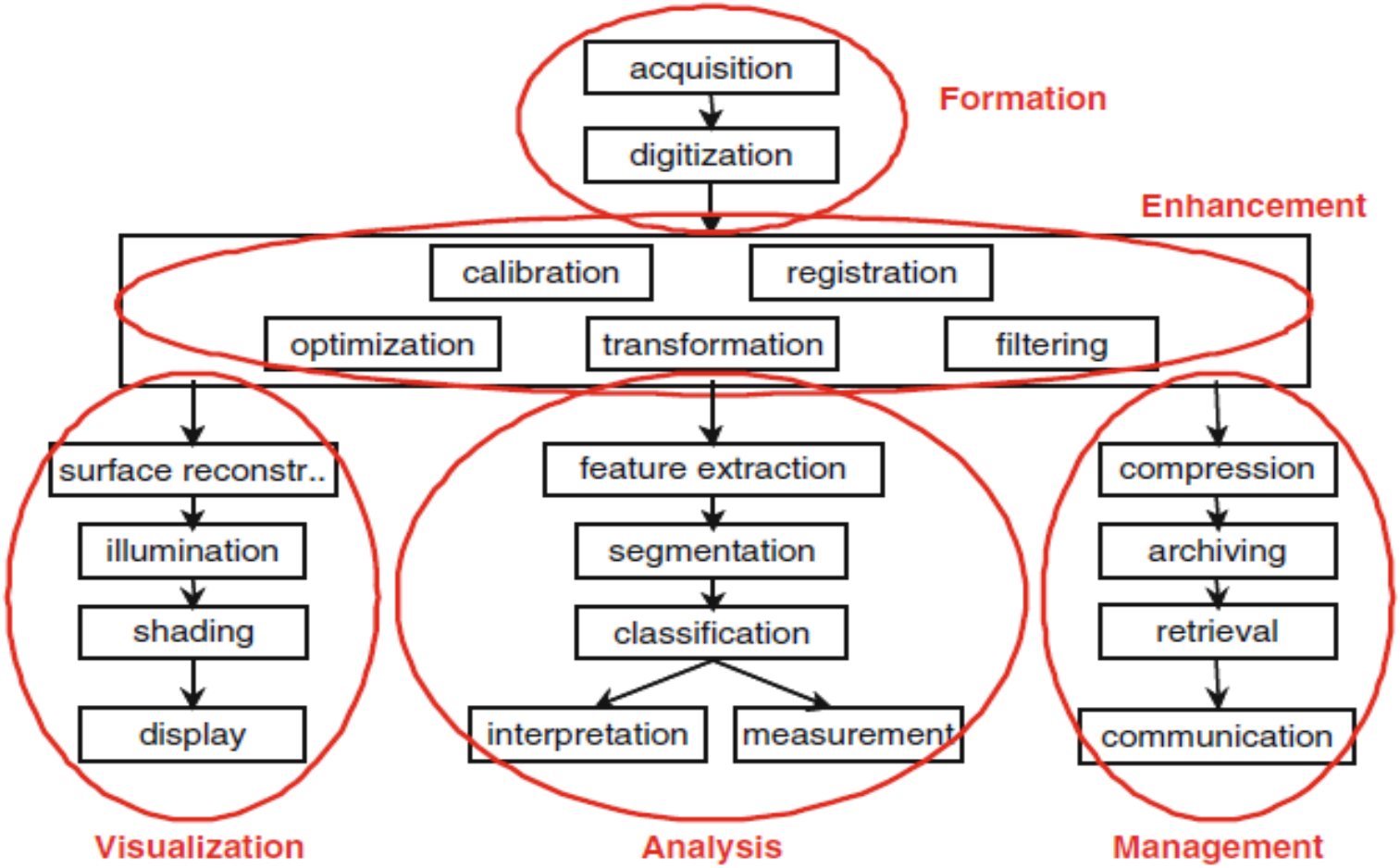


Image Formation

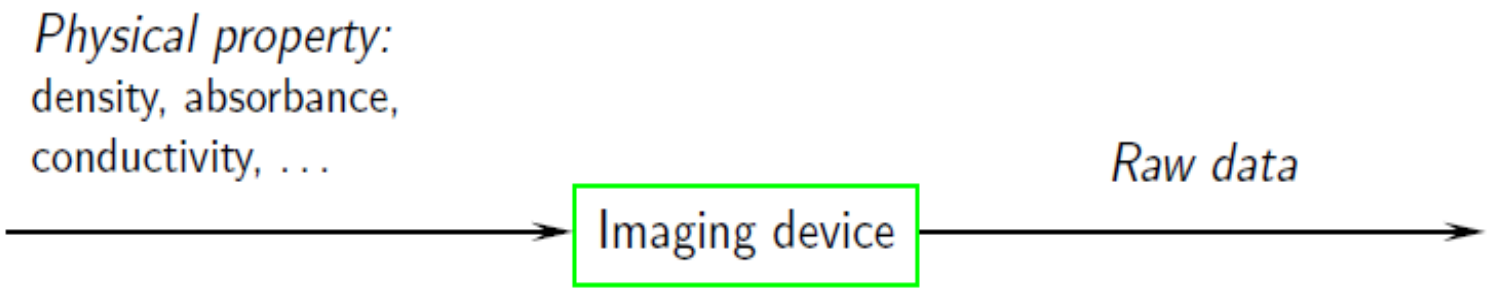
- It includes all the steps from capturing the image to forming a digital image matrix.
  - Image Acquisition
  - Digitization

Image Acquisition

- Imaging Database
- Imaging Devices

  Examples:

  - CT, X-rays, MRI, ultrasound, . . .
  - Body Scanner

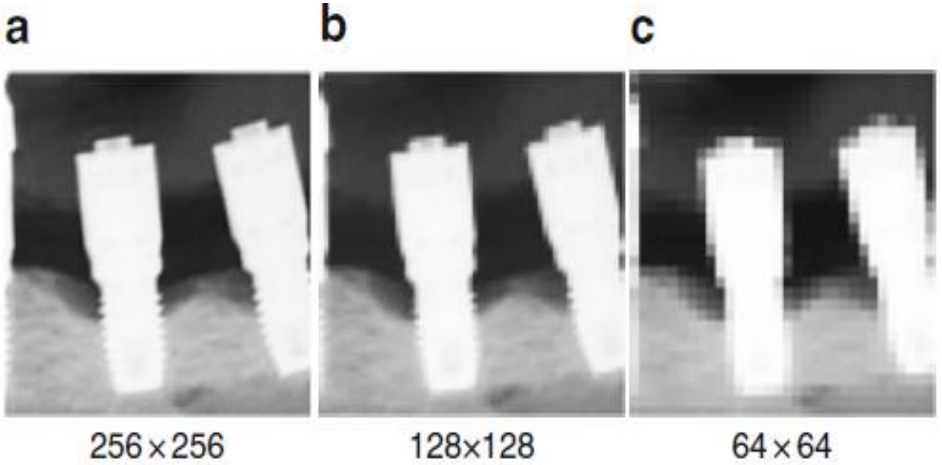


Digitization

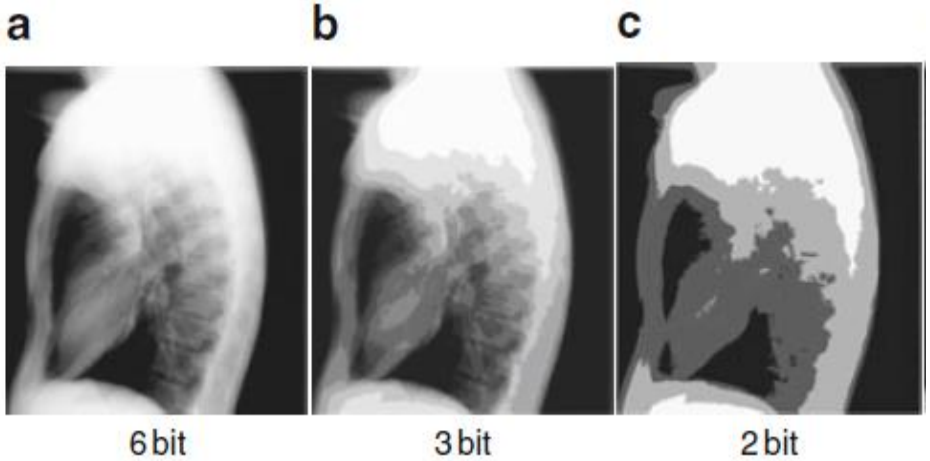
- Digital image processing implies a discrete nature of the images.
- There are two steps in which it is done:
  - Sampling
  - Quantization
- The sampling rate determines the spatial resolution of the digitized image
- The quantization level determines the number of grey levels in the digitized image.

Example
(using image processing techniques)

- Sampling
  - Resolution

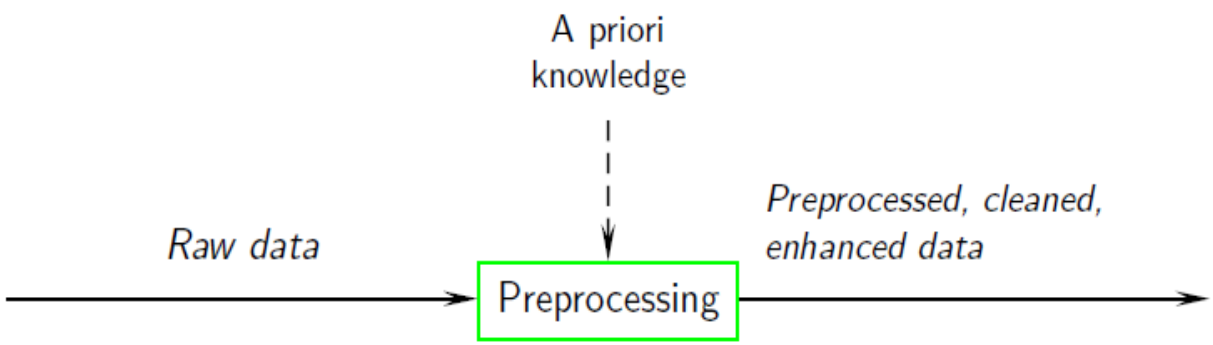


- Quantization
  - Grey Level



Preprocessing

- Examples:
- Noise suppression, contract enhancement, intensity equalization,
- outlier elimination, bias compensation, time/space filtering, . . .



Preprocessing

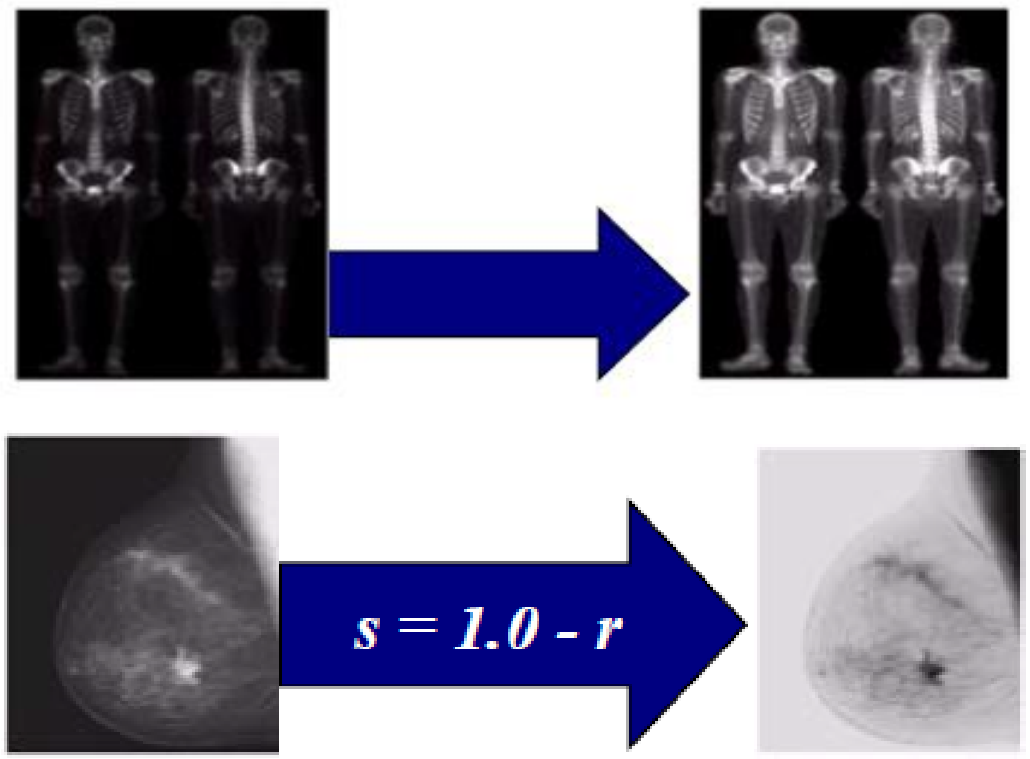
- Enhancement
- Filtering
- Registration
- Calibration
- Transformation

Enhancement

- The objectives of image enhancement techniques is to process an image so that the result is more suitable than the original image for a specific application .
- Image enhancement techniques can be divided into two broad categories:
- 1.Spatial domain methods .
  - Point Processing
  - Histogram Equalization
  - Image Subtraction
- 2 Frequency domain methods.
  - Fourier transform

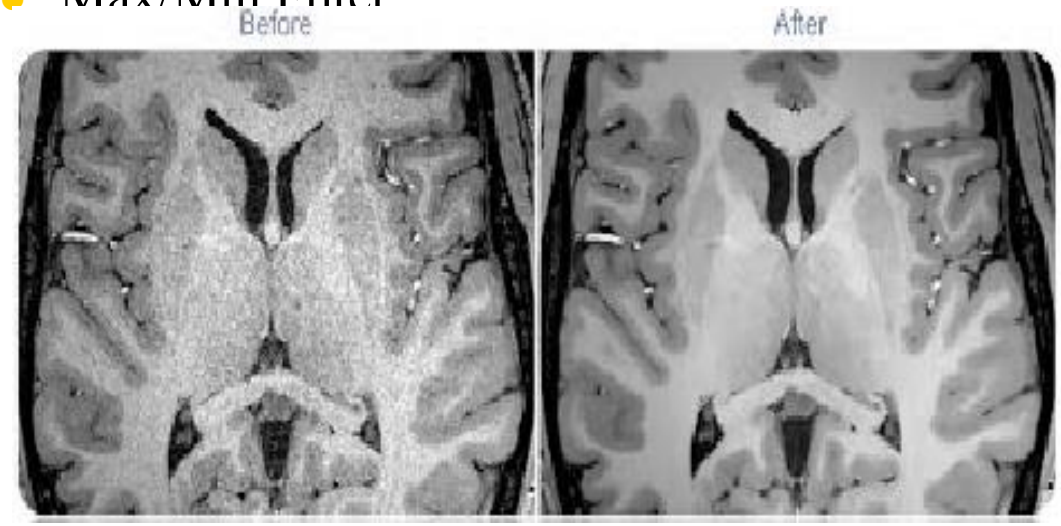
Enhancement

- Example



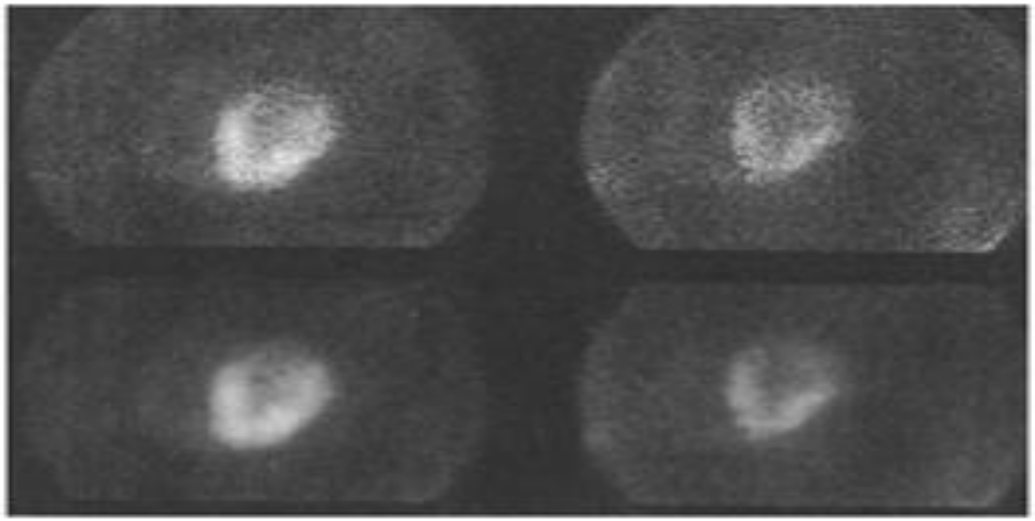
Filtering

- The objective of filtering is to remove noise.
- Techniques
  - Averaging Filter
  - Median Filter
  - Max/Min Filter



Smoothing

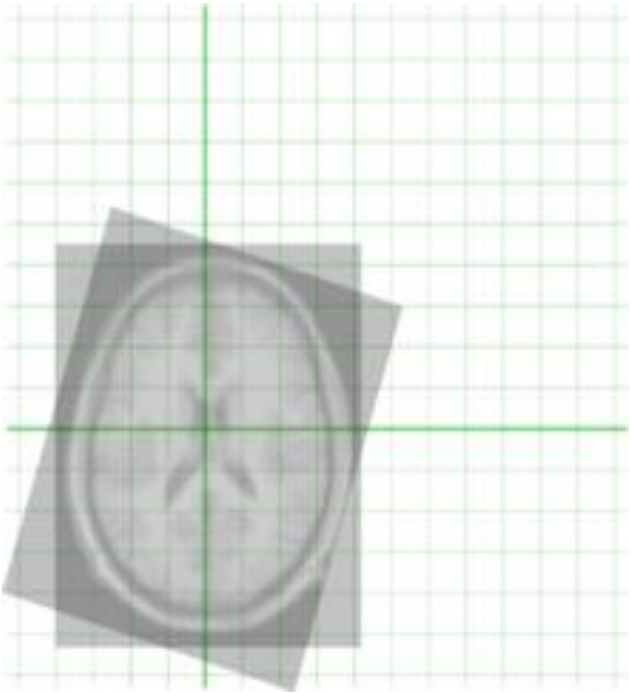
- The aim of image smoothing is to diminish the effects of camera noise, spurious pixel values, missing pixel values etc.
- Two methods used for image smoothing.
  - Neighborhood averaging and
  - Edge-preserving smoothing.



Registration

- Medical image registration, for data of the same patient taken at different points in time such as change detection or tumor monitoring.
- **Unimodal Registration**

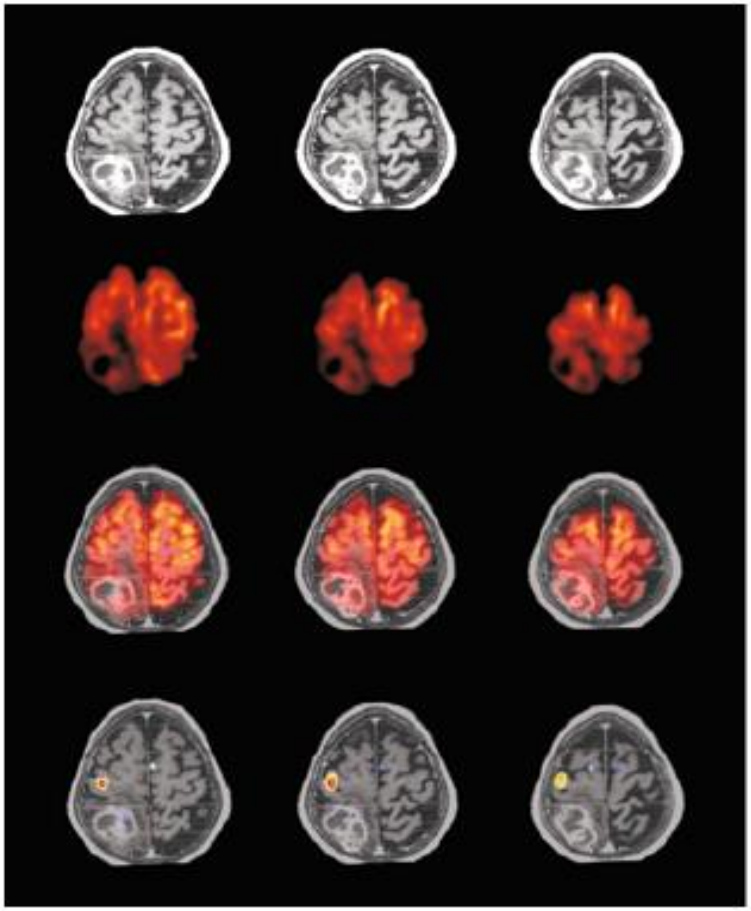
This term refers to the relative calibration of images that have been acquired with the same modality.



Registration

Multi-Modal Registration

The images to be compared are captured with different modalities.

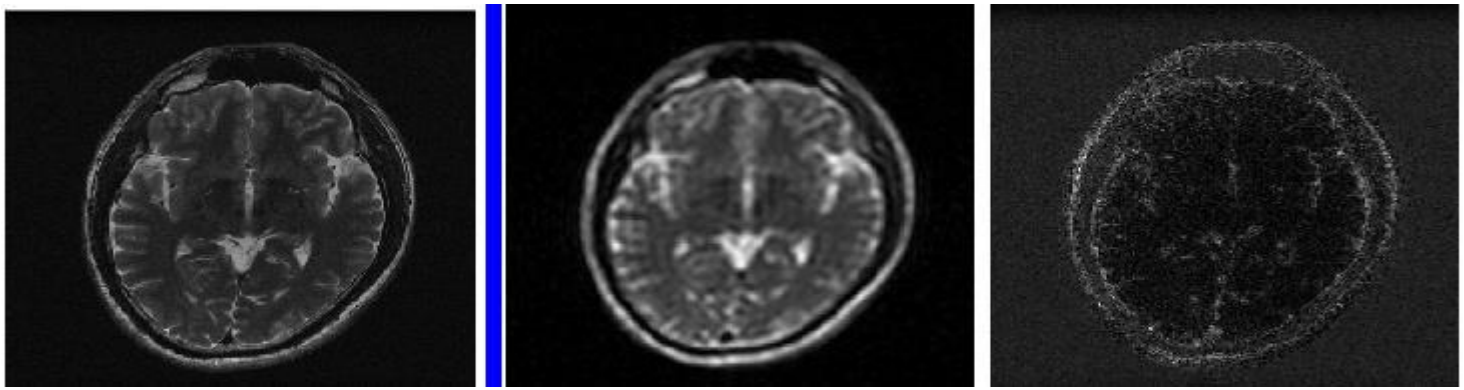


Transformation

- Image transforms can be simple arithmetic operations on images or complex mathematical operations which convert images from one representation to another.

- The transformation is intended to select the most prominent or relevant features.
  - Discrete Fourier Transform DFT
  - Discrete Cosine transform DCT
  - Discrete Wavelet transform DWT

Example



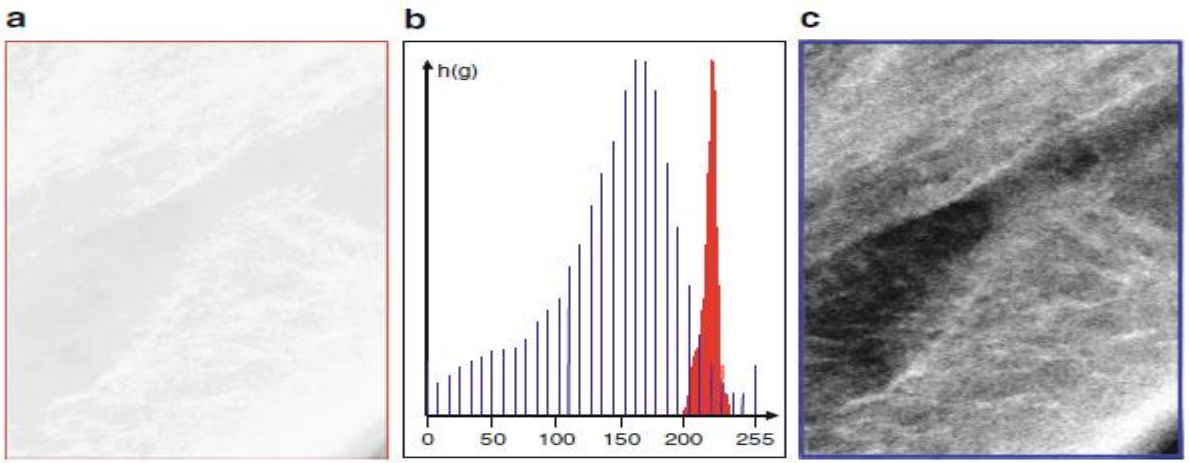
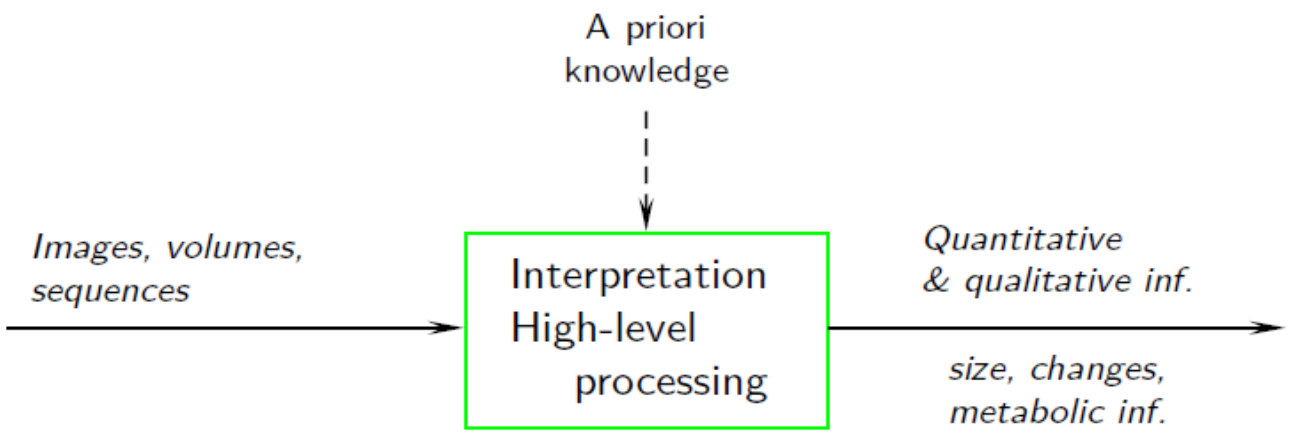


Image analysis

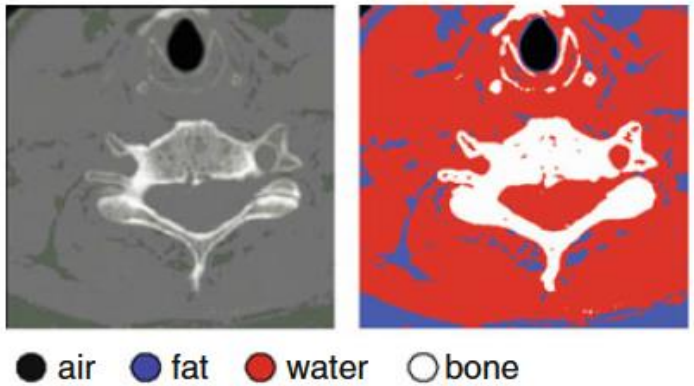
- It includes all the steps of processing
- Like,
  - Feature Extraction
  - Segmentation

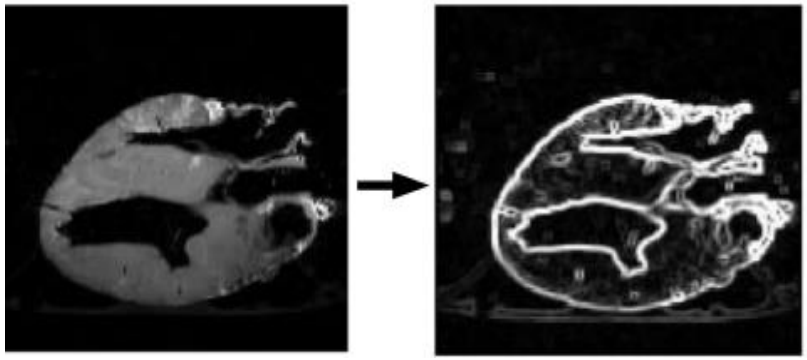


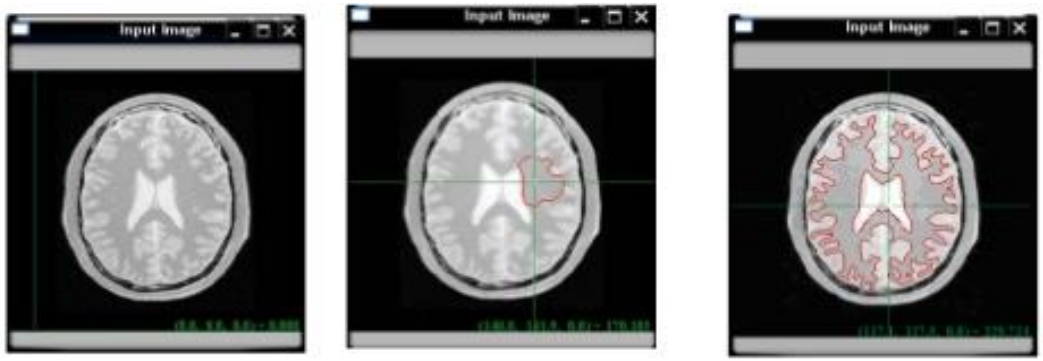
Segmentation

- Pixel based Segmentation
  - Thresolding
  - Adaptive Thresolding
  - Clustering
- Edge based Segmentation
  - Live wire Segmentation
  - Snake Algorithm
  - Active contour Algorithm
- Region Based Segmentation
  - Agglomerative Algorithm
  - Divisive Algorithm
- Hybrid Algorithm

Segmentation







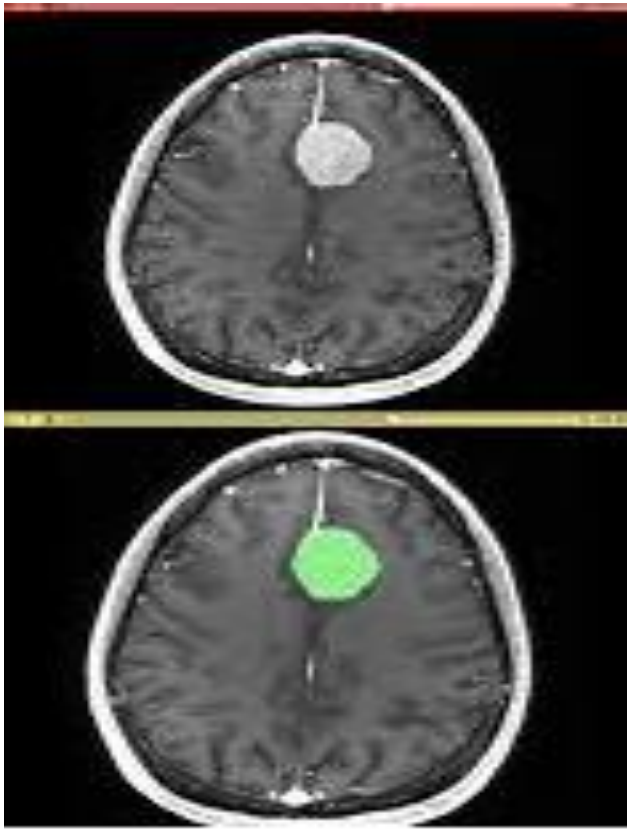
Original image *Initial segmentation* *Final segmentation*

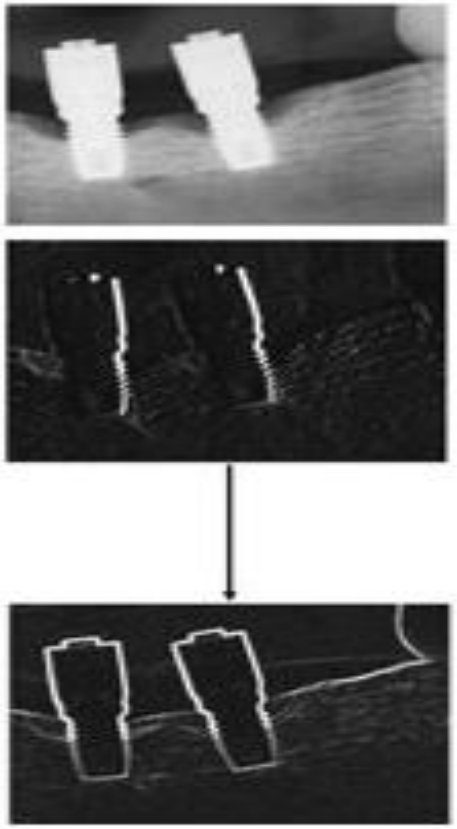
Feature extraction

- Feature extraction is defined as the first stage of intelligent (high level) image analysis.

- It is followed by segmentation and classification
- Levels Of Feature Extraction:
  - Data level
  - Region Level
  - Texture Level
  - Edge Level
  - Pixel Level

Example





Classification

- Statistic Classifiers
- Syntactic Classifiers
- Computational Intelligence-Based Classifiers
  - Neural network
  - Fuzzy Algorithm

Classification

- Comparison to developed atlases
- Use of knowledge databases
- Classify as normal/abnormal (brain)
- Classify lung nodules as benign/malignant
- Determine cancer/non-cancer

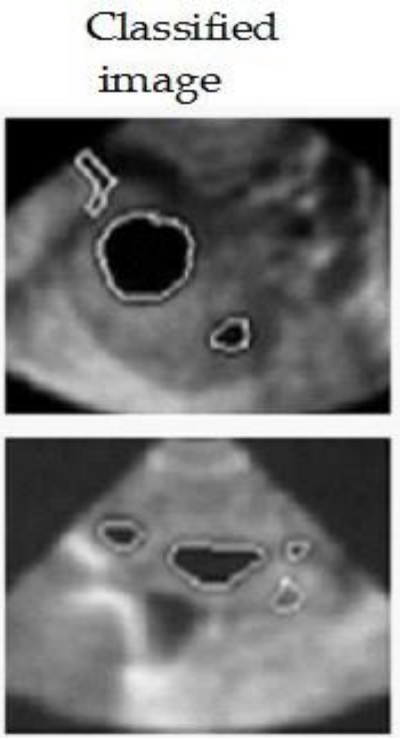
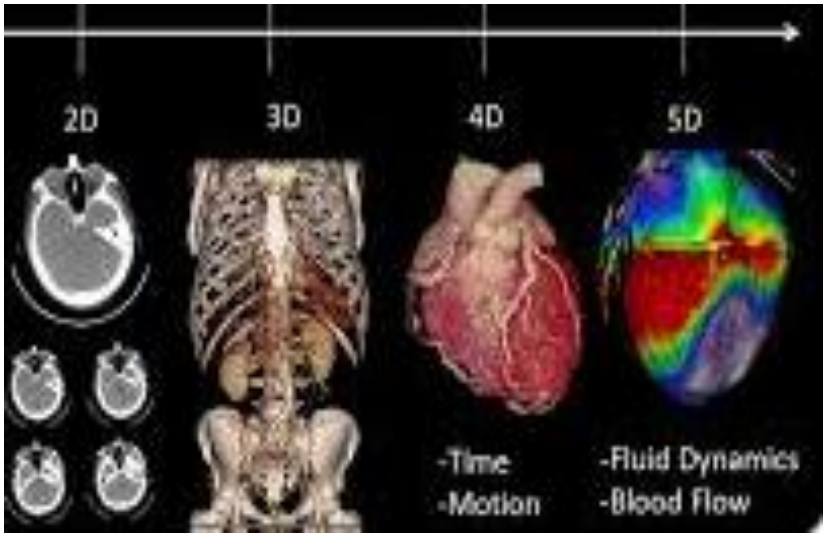


Image Visualization

- It refers to all types of manipulation of this matrix, resulting in an optimized output of the image.

- It includes,
  - Shading
  - Display
  - Reconstruction



Example

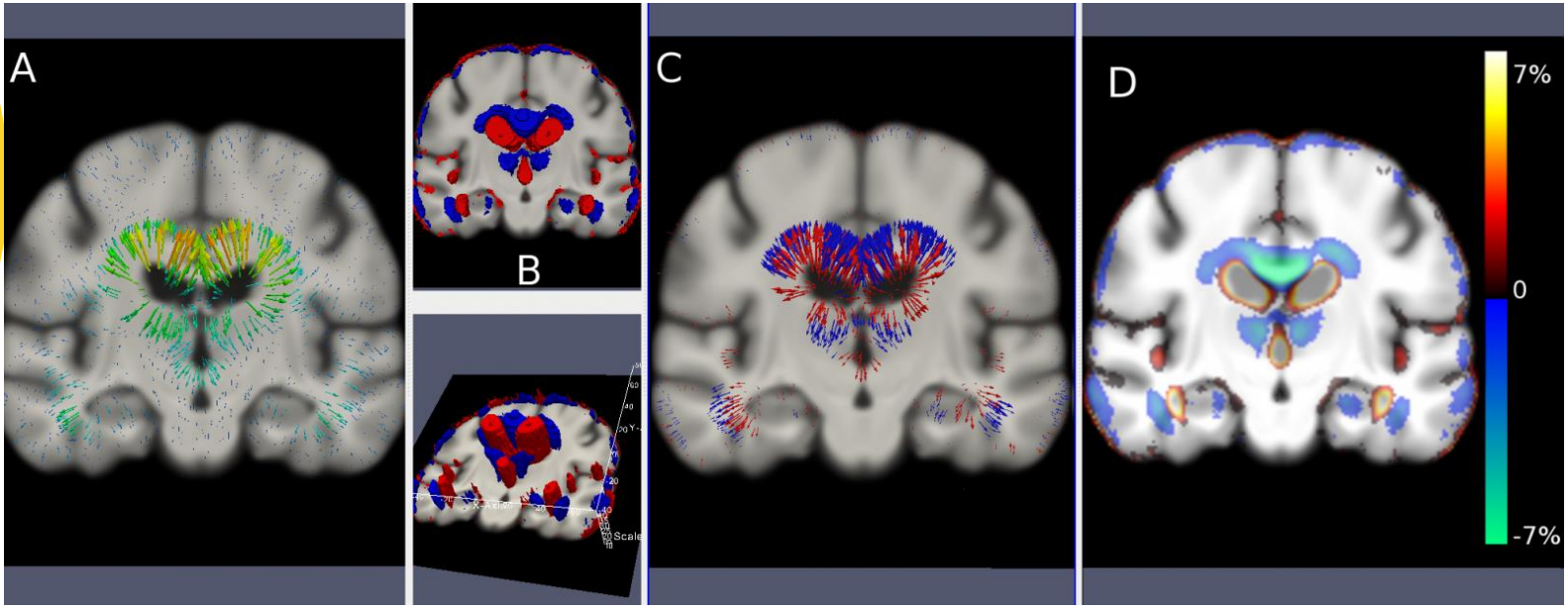


Image Management

- It sums up all techniques that provide the efficient storage, communication, transmission, archiving, and access (retrieval) of image data. Thus, the methods of telemedicine are also a part of the image management.

- It Includes,
  - Compression
  - Archive
  - Retrieval
  - Communication

Image Management

- Compression
- Retrieval
  - Content Based Retrieval
  - Text Based Retrieval
- Communication
  - DCOM
- Archiving
  - PACS Architecture

General Performance Measures

- Positive: Object was observed
- Negative: Object was not observed
- True Positive
- False Negative
- True Negative
- False Positive

General Performance Measures

True Condition

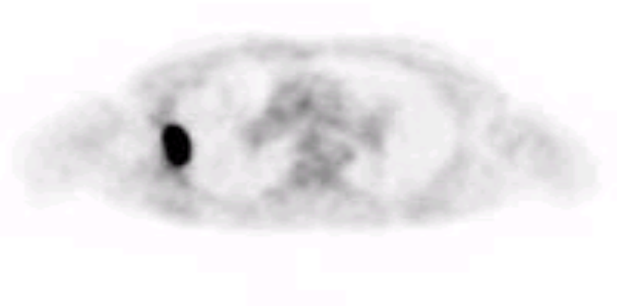
Observed Information	Object is present.	Object is NOT present.
Object is observed.	**True Positive**	**False Positive**
Object is NOT observed.	**False Negative**	**True Negative**

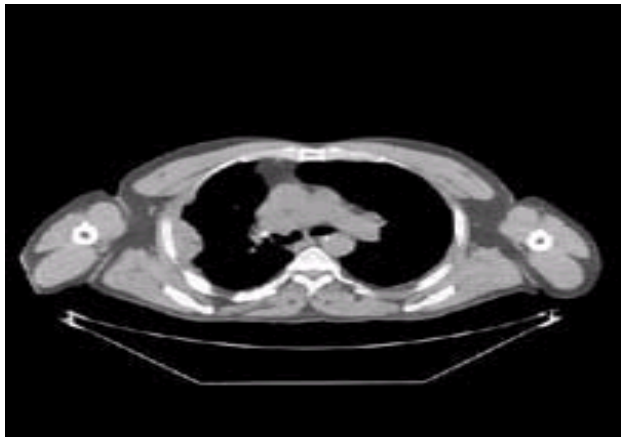
Fusion Imaging

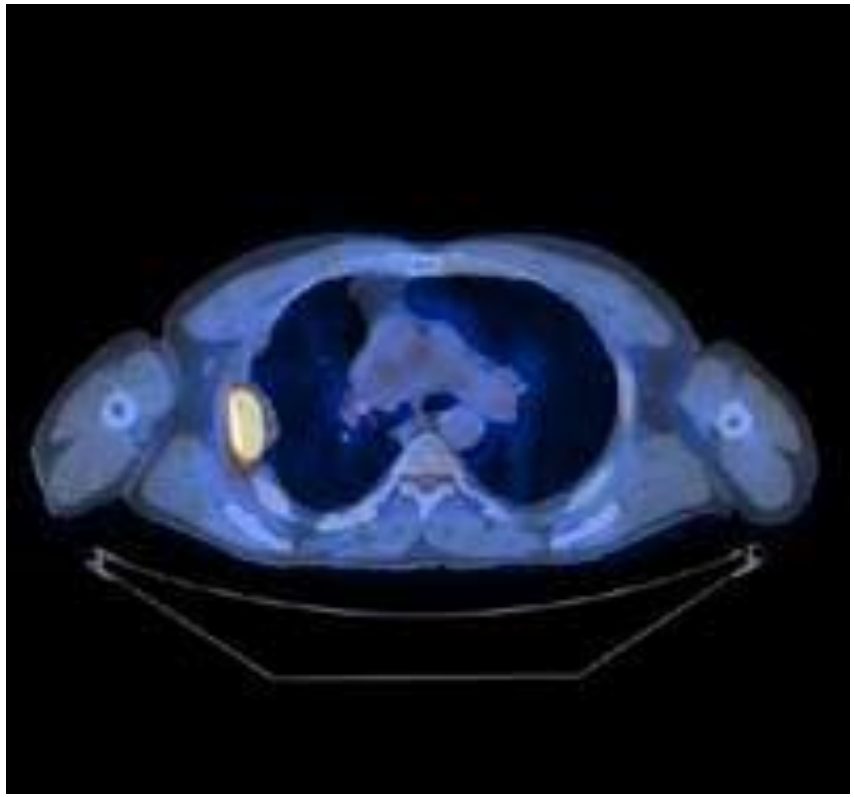
- Melting together images from different modalities to create a new (hybrid) image.
- **Multimodal Fusion**
  - Images of different modalities: PET, CT,MRI etc.

Example

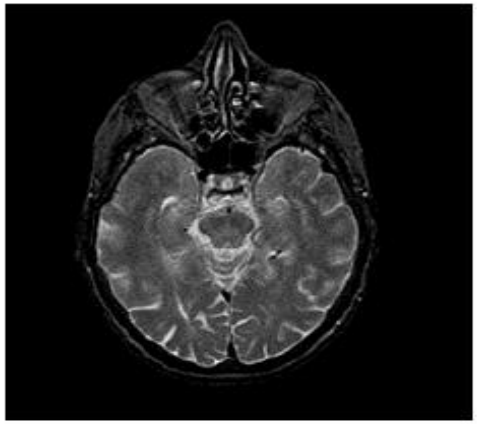
CT and
PET

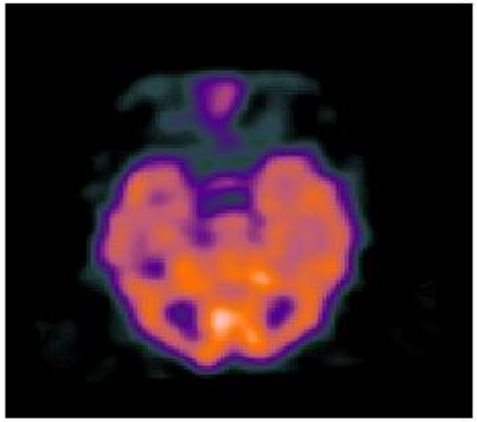




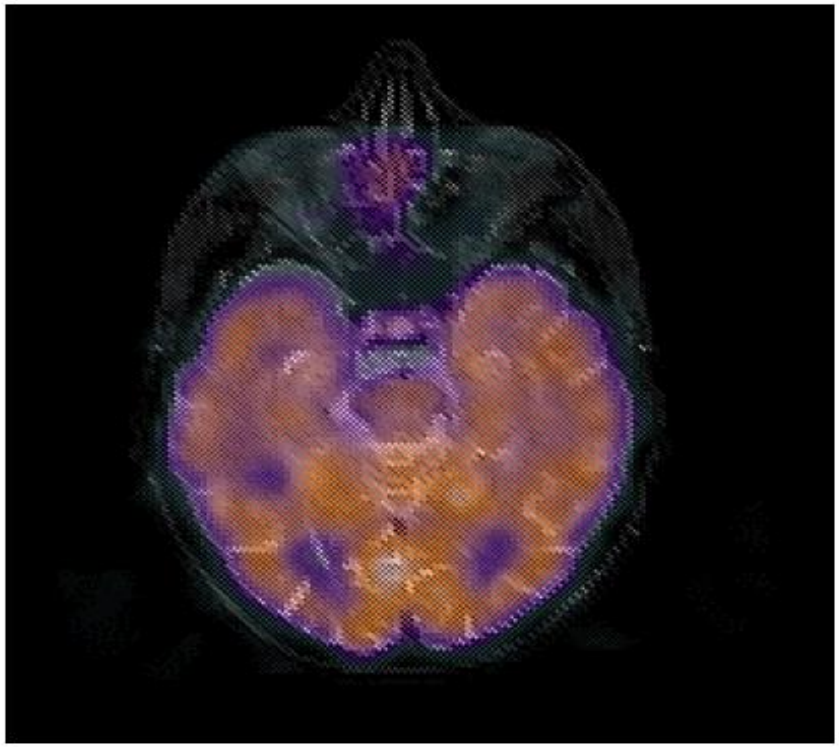


Example





NMR + SPECT



List of Applications

- Diagnosis
  - Combining information from multiple imaging modalities
- Studying disease progression
  - Monitoring changes in size, shape, position or image intensity over time
- Image guided surgery or radiotherapy
  - Relating pre-operative images and surgical plans to the physical reality of the patient
- Patient comparison or atlas construction
  - Relating one individual's anatomy to a standardized atlas

Conclusion

- The pragmatic issues of image generation, processing, presentation, and archiving stood in the focus of research in image processing.
- Because available computers at that time had by far not the necessary capacity to hold and modify large image data in memory.
- The former computation speed of image processing allowed only offline calculations.
- Until today, the automatic interpretation of images still is a major goal.
- Segmentation, classification, and measurements of biomedical images is continuously improved and validated more accurately, since validation is based on larger studies with high volumes of data.

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